

7.3 Quadratic Forms

In this section we use matrix methods to study real-valued functions of several variables in which each term is either the square of a variable or the product of two variables. Such functions arise in geometry, vibrations of mechanical systems, statistics, and electrical engineering.

Definition of a Quadratic Form

Expressions of the form $a_1x_1 + a_2x_2 + \cdots + a_nx_n$ are *linear forms* on R^n . **Quadratic forms** on R^n involve squares and cross products of the variables. Terms $a_kx_ix_j$ with $i \neq j$ are called **cross product terms**; it is conventional to combine x_ix_j and x_jx_i so a general quadratic form on R^2 is

$$a_1x_1^2 + a_2x_2^2 + 2a_3x_1x_2 \quad (1)$$

and on R^3 ,

$$a_1x_1^2 + a_2x_2^2 + a_3x_3^2 + 2a_4x_1x_2 + 2a_5x_1x_3 + 2a_6x_2x_3 \quad (2)$$

If A is a symmetric $n \times n$ matrix and $\mathbf{x}$ is the $n \times 1$ column vector of variables, then (1) and (2) can be expressed in matrix form as $\mathbf{x}^T A \mathbf{x}$ (a 1×1 scalar). The diagonal entries of A are the coefficients of the squared terms, and the off-diagonal entries are half the coefficients of the cross product terms. We define

$$Q_A(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} \quad (3)$$

as the **quadratic form associated with A** . In dot product notation:

$$\boxed{\mathbf{x}^T A \mathbf{x} = \mathbf{x} \cdot A \mathbf{x} = A \mathbf{x} \cdot \mathbf{x}} \quad (4)$$

When A is diagonal with entries $\lambda_1, \dots, \lambda_n$, there are no cross product terms:

$$\mathbf{x}^T A \mathbf{x} = [x_1 \ \cdots \ x_n] \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \lambda_1 x_1^2 + \lambda_2 x_2^2 + \cdots + \lambda_n x_n^2$$

EXAMPLE 1 | Expressing Quadratic Forms in Matrix Notation

Express each QF as $\mathbf{x}^T A \mathbf{x}$ with A symmetric. (a) $2x^2 + 6xy - 5y^2$ (b) $x_1^2 + 7x_2^2 - 3x_3^2 + 4x_1x_2 - 2x_1x_3 + 8x_2x_3$

SOLUTION:

$$2x^2 + 6xy - 5y^2 = [x \ y] \begin{bmatrix} 2 & 3 \\ 3 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x_1^2 + 7x_2^2 - 3x_3^2 + 4x_1x_2 - 2x_1x_3 + 8x_2x_3 = [x_1 \ x_2 \ x_3] \begin{bmatrix} 1 & 2 & -1 \\ 2 & 7 & 4 \\ -1 & 4 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Change of Variable in a Quadratic Form

Three important problems involving quadratic forms:

Problem 1	If $\mathbf{x}^T A \mathbf{x}$ is a QF on R^2 or R^3 , what kind of curve or surface does $\mathbf{x}^T A \mathbf{x} = k$ represent?
Problem 2	If $\mathbf{x}^T A \mathbf{x}$ is a QF on R^n , what conditions on A make $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$?
Problem 3	If $\mathbf{x}^T A \mathbf{x}$ is a QF on R^n , what are the maximum and minimum values subject to $\ \mathbf{x}\ = 1$?

We treat Problems 1 and 2 here; Problem 3 is in the next section. The main tool is the substitution

$$\mathbf{x} = P\mathbf{y} \quad (5)$$

called a **change of variable**; if P is orthogonal, it is called an **orthogonal change of variable**.

Theorem 7.3.1 — The Principal Axes Theorem

If A is a symmetric $n \times n$ matrix, there is an orthogonal change of variable $\mathbf{x} = P\mathbf{y}$ that transforms $\mathbf{x}^T A \mathbf{x}$ into a quadratic form with no cross product terms:

$$\mathbf{x}^T A \mathbf{x} = \mathbf{y}^T D \mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A corresponding to the successive columns of P .

Proof. Substituting $\mathbf{x} = P\mathbf{y}$:

$$\mathbf{x}^T A \mathbf{x} = (P\mathbf{y})^T A (P\mathbf{y}) = \mathbf{y}^T P^T A P \mathbf{y} = \mathbf{y}^T (P^T A P) \mathbf{y} \quad (6)$$

Choose P to orthogonally diagonalize A , so $D = P^T A P$ is diagonal with eigenvalues on the diagonal. Then

$$\mathbf{x}^T A \mathbf{x} = \mathbf{y}^T D \mathbf{y} = [y_1 \ \cdots \ y_n] \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 \quad \blacksquare$$

EXAMPLE 2 | An Illustration of the Principal Axes Theorem

Find an orthogonal change of variable that eliminates the cross product terms in $Q = x_1^2 - x_3^2 - 4x_1x_2 + 4x_2x_3$, and express Q in terms of the new variables.

SOLUTION: The quadratic form in matrix notation is

$$Q = \mathbf{x}^T A \mathbf{x}, \quad A = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 0 & 2 \\ 0 & 2 & -1 \end{bmatrix}$$

The characteristic equation of A is

$$\begin{vmatrix} \lambda - 1 & 2 & 0 \\ 2 & \lambda & -2 \\ 0 & -2 & \lambda + 1 \end{vmatrix} = \lambda^3 - 9\lambda = \lambda(\lambda + 3)(\lambda - 3) = 0$$

so the eigenvalues are $\lambda = 0, -3, 3$. Orthonormal bases for the three eigenspaces are:

$$\lambda = 0 : \begin{bmatrix} 2/3 \\ 1/3 \\ 2/3 \end{bmatrix}, \quad \lambda = -3 : \begin{bmatrix} -1/3 \\ -2/3 \\ 2/3 \end{bmatrix}, \quad \lambda = 3 : \begin{bmatrix} -2/3 \\ 2/3 \\ 1/3 \end{bmatrix}$$

The substitution $\mathbf{x} = P\mathbf{y}$ that eliminates cross product terms is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2/3 & -1/3 & -2/3 \\ 1/3 & -2/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

This produces the new quadratic form

$$Q = \mathbf{y}^T (P^T A P) \mathbf{y} = [y_1 \ y_2 \ y_3] \begin{bmatrix} 0 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = -3y_2^2 + 3y_3^2$$

in which there are no cross product terms.

Remark. An orthogonal change of variable does not alter the range of a quadratic form: the set of all values of $\mathbf{x}^T A \mathbf{x}$ as $\mathbf{x}$ varies over R^n equals the set of all values of $\mathbf{y}^T (P^T A P) \mathbf{y}$ as $\mathbf{y}$ varies over R^n .

Quadratic Forms in Geometry

A **conic section** (or **conic**) is a curve formed by intersecting a double-napped cone with a plane. The most important types are ellipses, parabolas, and hyperbolas; if the cutting plane passes through the vertex the intersection is called a **degenerate conic** (a point, a pair of intersecting lines, or a single line).

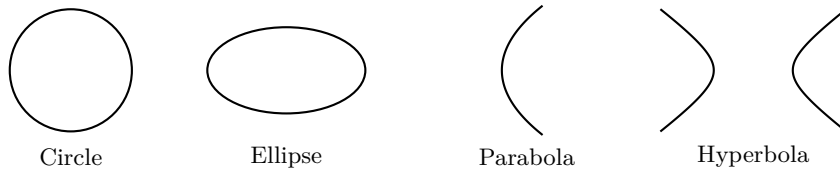


Figure 1: The four main types of conic sections.

Quadratic forms in R^2 arise naturally in the study of conics. An equation of the form

$$ax^2 + 2bxy + cy^2 + dx + ey + f = 0 \tag{7}$$

(with a, b, c not all zero) represents a conic section. If $d = e = 0$, Equation (7) becomes

$$ax^2 + 2bxy + cy^2 + f = 0 \tag{8}$$

a **central conic** (includes circles, ellipses, and hyperbolas, but not parabolas). If further $b = 0$:

$$ax^2 + cy^2 + f = 0 \quad (9)$$

this is a **central conic in standard position**. The most important types are:

Type	Equation	Conditions
Ellipse (major axis along x)	$\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1$	$\alpha \geq \beta > 0$
Ellipse (major axis along y)	$\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1$	$\beta \geq \alpha > 0$
Hyperbola (opens left/right)	$\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$	$\alpha > 0, \beta > 0$
Hyperbola (opens up/down)	$\frac{y^2}{\beta^2} - \frac{x^2}{\alpha^2} = 1$	$\alpha > 0, \beta > 0$

Setting $k = -f$ and expressing (8) and (9) in matrix form:

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = k \quad \text{and} \quad \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = k \quad (10)$$

A cross product term ($b \neq 0$) signals that the graph is rotated out of standard position. The three-dimensional analogs are:

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = k \quad \text{and} \quad \begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = k \quad (11)$$

The graphs of these in R^3 (when a, b, c not all zero) are called **central quadrics**; the second is a **central quadric in standard position**.

Identifying Conic Sections

A central conic $ax^2 + 2bxy + cy^2 = k$ is in standard position if $b = 0$. When $b \neq 0$, we can identify the conic by writing the equation as

$$\mathbf{x}^T A \mathbf{x} = k, \quad A = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \quad (13)$$

and finding an orthogonal change of variable $\mathbf{x} = P\mathbf{x}'$ with $\det(P) = 1$ (a rotation) that diagonalizes A . In the rotated $x'y'$ -coordinate system, (13) becomes

$$\mathbf{x}'^T D \mathbf{x}' = k, \quad \text{i.e.,} \quad \lambda_1 x'^2 + \lambda_2 y'^2 = k \quad (17)$$

where λ_1, λ_2 are the eigenvalues of A . The first column of P (a unit eigenvector for λ_1) gives the direction of the positive x' -axis. A useful formula: when $b \neq 0$, the rotation angle θ satisfies

$$\boxed{\cot 2\theta = \frac{a - c}{2b}} \quad (19)$$

EXAMPLE 3 | Identifying a Conic by Eliminating the Cross Product Term

- (a) Identify the conic $5x^2 - 4xy + 8y^2 - 36 = 0$ by rotating the xy -axes to standard position.
(b) Find the angle θ of rotation.

SOLUTION (a): Write as $\mathbf{x}^T A \mathbf{x} = 36$ with

$$A = \begin{bmatrix} 5 & -2 \\ -2 & 8 \end{bmatrix}$$

The characteristic polynomial of A is

$$\begin{vmatrix} \lambda - 5 & 2 \\ 2 & \lambda - 8 \end{vmatrix} = (\lambda - 4)(\lambda - 9)$$

so the eigenvalues are $\lambda = 4$ and $\lambda = 9$. Orthonormal bases for the eigenspaces are:

$$\lambda = 4: \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}, \quad \lambda = 9: \begin{bmatrix} -1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

Thus A is orthogonally diagonalized by

$$P = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \quad (18)$$

Since $\det(P) = 1$, the substitution $\mathbf{x} = P\mathbf{x}'$ is a rotation. In the $x'y'$ -system:

$$\begin{bmatrix} x' & y' \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} = 36 \Rightarrow 4x'^2 + 9y'^2 = 36 \Rightarrow \frac{x'^2}{9} + \frac{y'^2}{4} = 1$$

This is an ellipse with major axis of length $2\alpha = 6$ in the x' -direction and minor axis of length $2\beta = 4$ in the y' -direction.

SOLUTION (b): From (18), $\cos \theta = 2/\sqrt{5}$ and $\sin \theta = 1/\sqrt{5}$, so

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{1}{2}, \quad \theta = \tan^{-1} \frac{1}{2} \approx 26.6^\circ$$

Positive Definite Quadratic Forms

We now consider Problem 2: conditions under which $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Definition 1

A quadratic form $\mathbf{x}^T A \mathbf{x}$ (and the matrix A) is called:

- **positive definite** if $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$
- **negative definite** if $\mathbf{x}^T A \mathbf{x} < 0$ for all $\mathbf{x} \neq \mathbf{0}$
- **indefinite** if $\mathbf{x}^T A \mathbf{x}$ takes both positive and negative values

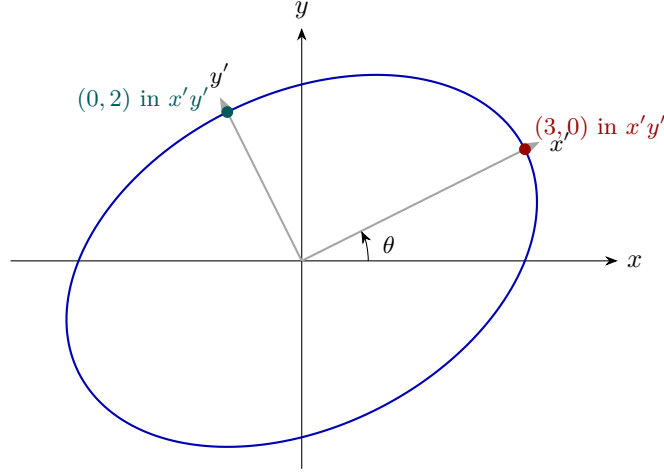


Figure 2: The ellipse $5x^2 - 4xy + 8y^2 = 36$ in the original xy -frame, with the rotated $x'y'$ -axes along the principal axes.

Theorem 7.3.2

If A is a symmetric matrix, then:

- (a) $\mathbf{x}^T A \mathbf{x}$ is positive definite iff all eigenvalues of A are positive.
- (b) $\mathbf{x}^T A \mathbf{x}$ is negative definite iff all eigenvalues of A are negative.
- (c) $\mathbf{x}^T A \mathbf{x}$ is indefinite iff A has at least one positive and at least one negative eigenvalue.

Proof. By Theorem 7.3.1 there is an orthogonal change of variable $\mathbf{x} = P\mathbf{y}$ with

$$\mathbf{x}^T A \mathbf{x} = \mathbf{y}^T D \mathbf{y} = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2 \quad (23)$$

Since P is invertible, $\mathbf{y} \neq \mathbf{0} \Leftrightarrow \mathbf{x} \neq \mathbf{0}$, and the set of values of $\mathbf{x}^T A \mathbf{x}$ over all $\mathbf{x}$ equals the set of values of $\mathbf{y}^T D \mathbf{y}$ over all $\mathbf{y}$.

(a) and (b): If all $\lambda_i > 0$, then (23) is positive for every $\mathbf{y} \neq \mathbf{0}$, so $\mathbf{x}^T A \mathbf{x} > 0$. Conversely, if some $\lambda_i \leq 0$, taking $\mathbf{y}$ with $y_i = 1$ and all other $y_j = 0$ gives $\mathbf{x}^T A \mathbf{x} = \lambda_i \leq 0$. Part (b) is analogous.

(c): If $\lambda_1 > 0$ and $\lambda_2 < 0$, setting $y_1 = 1, y_2 = \cdots = 0$ gives $\mathbf{x}^T A \mathbf{x} = \lambda_1 > 0$, and setting $y_2 = 1, y_1 = y_3 = \cdots = 0$ gives $\mathbf{x}^T A \mathbf{x} = \lambda_2 < 0$. Conversely, if both signs occur in (23), at least one λ_i must be positive and one negative. ■

Remark. Two additional classifications:

- $\mathbf{x}^T A \mathbf{x}$ is **positive semidefinite** if $\mathbf{x}^T A \mathbf{x} \geq 0$ for $\mathbf{x} \neq \mathbf{0}$ (all eigenvalues ≥ 0)
- $\mathbf{x}^T A \mathbf{x}$ is **negative semidefinite** if $\mathbf{x}^T A \mathbf{x} \leq 0$ for $\mathbf{x} \neq \mathbf{0}$ (all eigenvalues ≤ 0)

EXAMPLE 4 | Positive Definite Quadratic Forms

Determine whether $\mathbf{x}^T A \mathbf{x}$ is positive definite for $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$.

SOLUTION: The entries are nonnegative, but the eigenvalues are $\lambda = 1, 4, -2$ (verify). Since A has

a negative eigenvalue, $\mathbf{x}^T A \mathbf{x}$ is indefinite. Writing out the QF:

$$\mathbf{x}^T A \mathbf{x} = 3x_1^2 + 2x_1x_2 + 2x_1x_3 + 4x_2x_3$$

Indeed, $\mathbf{x}^T A \mathbf{x} = 4$ for $(x_1, x_2, x_3) = (0, 1, 1)$ and $\mathbf{x}^T A \mathbf{x} = -4$ for $(0, 1, -1)$.

Classifying Conic Sections Using Eigenvalues

If $\mathbf{x}^T B \mathbf{x} = k$ with $k \neq 0$, set $A = (1/k)B$ and rewrite as

$$\mathbf{x}^T A \mathbf{x} = 1 \quad (20)$$

After rotating to eliminate cross product terms, (20) becomes

$$\lambda_1 x'^2 + \lambda_2 y'^2 = 1 \quad (21)$$

where λ_1, λ_2 are eigenvalues of A :

- Ellipse if $\lambda_1 > 0$ and $\lambda_2 > 0$
- No graph if $\lambda_1 < 0$ and $\lambda_2 < 0$
- Hyperbola if λ_1, λ_2 have opposite signs

For the ellipse case, (21) can be written

$$\frac{x'^2}{(1/\sqrt{\lambda_1})^2} + \frac{y'^2}{(1/\sqrt{\lambda_2})^2} = 1 \quad (22)$$

so the axes have lengths $2/\sqrt{\lambda_1}$ and $2/\sqrt{\lambda_2}$.

Theorem 7.3.3

If A is a symmetric 2×2 matrix, then:

- (a) $\mathbf{x}^T A \mathbf{x} = 1$ represents an ellipse if A is positive definite.
- (b) $\mathbf{x}^T A \mathbf{x} = 1$ has no graph if A is negative definite.
- (c) $\mathbf{x}^T A \mathbf{x} = 1$ represents a hyperbola if A is indefinite.

Identifying Positive Definite Matrices

The k th **principal submatrix** of an $n \times n$ matrix A is the $k \times k$ submatrix formed by the first k rows and columns of A . For a 4×4 matrix these are:

$$\begin{bmatrix} a_{11} \end{bmatrix}, \quad \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad A$$

Theorem 7.3.4

If A is a symmetric matrix, then:

- (a) A is positive definite iff the determinant of every principal submatrix is positive.
- (b) A is negative definite iff the determinants of the principal submatrices alternate in sign, beginning with a negative value.
- (c) A is indefinite iff it is neither positive nor negative definite and at least one principal submatrix has a positive determinant and at least one has a negative determinant.

EXAMPLE 5 | Working with Principal Submatrices

The matrix

$$A = \begin{bmatrix} 2 & -1 & -3 \\ -1 & 2 & 4 \\ -3 & 4 & 9 \end{bmatrix}$$

is positive definite since its principal submatrix determinants

$$|2| = 2, \quad \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3, \quad \begin{vmatrix} 2 & -1 & -3 \\ -1 & 2 & 4 \\ -3 & 4 & 9 \end{vmatrix} = 1$$

are all positive. Thus all eigenvalues of A are positive and $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Exercise Set 7.3

In Exercises 1–2, express the quadratic form in the matrix notation $\mathbf{x}^T A \mathbf{x}$, where A is symmetric.

1. **a.** $3x_1^2 + 7x_2^2$ **b.** $4x_1^2 - 9x_2^2 - 6x_1x_2$
c. $9x_1^2 - x_2^2 + 4x_3^2 + 6x_1x_2 - 8x_1x_3 + x_2x_3$

In Exercises 3–4, find a formula for the quadratic form that does not use matrices.

$$3. \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -3 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

In Exercises 5–8, find an orthogonal change of variables that eliminates the cross product terms in Q , and express Q in terms of the new variables.

5. $Q = 2x_1^2 + 2x_2^2 - 2x_1x_2$
7. $Q = 3x_1^2 + 4x_2^2 + 5x_3^2 + 4x_1x_2 - 4x_2x_3$

In Exercises 9–10, express the quadratic equation in the matrix form $\mathbf{x}^T A \mathbf{x} + K \mathbf{x} + f = 0$, where $\mathbf{x}^T A \mathbf{x}$ is the associated quadratic form and K is an appropriate matrix.

9. **a.** $2x^2 + xy + x - 6y + 2 = 0$ **b.** $y^2 + 7x - 8y - 5 = 0$

In Exercises 11–12, identify the conic section represented by the equation.

11. **a.** $2x^2 + 5y^2 = 20$ **b.** $x^2 - y^2 - 8 = 0$ **c.** $7y^2 - 2x = 0$

In Exercises 13–16, identify the conic section by rotating axes to place it in standard position. Find the equation in the rotated coordinates and the angle of rotation.

13. $2x^2 - 4xy - y^2 + 8 = 0$

16. $x^2 + xy + y^2 = \frac{1}{2}$

In Exercises 17–18, determine by inspection whether the matrix is positive definite, negative definite, indefinite, positive semidefinite, or negative semidefinite.

17. **a.** $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ **b.** $\begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$ **c.** $\begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$ **d.** $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ **e.** $\begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix}$

In Exercises 19–24, classify the quadratic form as positive definite, negative definite, indefinite, positive semidefinite, or negative semidefinite.

19. $x_1^2 + x_2^2$

22. $-(x_1 - x_2)^2$

In Exercises 25–26, show that A is positive definite first by using Theorem 7.3.2 and then by using Theorem 7.3.4.

25. **a.** $A = \begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$ **b.** $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

26. **a.** $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ **b.** $A = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

In Exercises 27–28, use Theorem 7.3.4 to classify the matrix as positive definite, negative definite, or indefinite.

27. **a.** $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & -1 & 3 \\ 2 & 3 & 2 \end{bmatrix}$ **b.** $A = \begin{bmatrix} -3 & 2 & 0 \\ 2 & -3 & 0 \\ 0 & 0 & -5 \end{bmatrix}$

In Exercises 29–30, find all values of k for which the quadratic form is positive definite.

29. $5x_1^2 + x_2^2 + kx_3^2 + 4x_1x_2 - 2x_1x_3 - 2x_2x_3$

Working with Proofs

31. Let $\mathbf{x}^T A \mathbf{x}$ be a quadratic form in $x_1, \dots, x_n$, and define $T: R^n \rightarrow R$ by $T(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$.

a. Show that $T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + 2\mathbf{x}^T A \mathbf{y} + T(\mathbf{y})$.

b. Show that $T(c\mathbf{x}) = c^2 T(\mathbf{x})$.

33. In statistics, the *sample mean* and *sample variance* of $\mathbf{x} = (x_1, \dots, x_n)$ are

$$\bar{x} = \frac{1}{n}(x_1 + \dots + x_n), \quad s_x^2 = \frac{1}{n-1}[(x_1 - \bar{x})^2 + \dots + (x_n - \bar{x})^2]$$

a. Express s_x^2 in matrix notation $\mathbf{x}^T A \mathbf{x}$ where A is symmetric.

b. Is s_x^2 a positive definite quadratic form? Explain.

36. Prove: if $b \neq 0$, the cross product term in $ax^2 + 2bxy + cy^2 = k$ can be eliminated by rotating through an angle θ satisfying $\cot 2\theta = \frac{a-c}{2b}$.